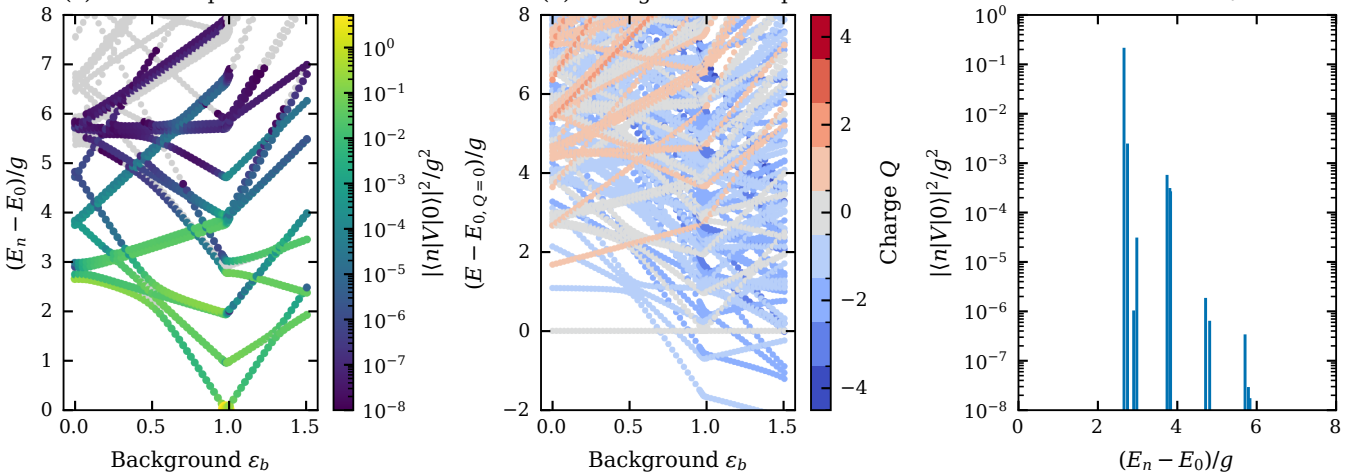

(a) Neutral spectrum
$(E_n - E_0)/g$
Background ε_b
$|\langle n|V|0\rangle|^2/g^2$
(b) Charge-resolved spectrum
$(E - E_{0,Q=0})/g$
Background ε_b
Charge Q
(c) Transitions at $\varepsilon_b = 0$
$|\langle n|V|0\rangle|^2/g^2$
$(E_n - E_0)/g$